



PITCH CLOCK IN PLAY

EXPLORING THE IMPACT OF THE INTRODUCTION
OF THE PITCH CLOCK ON MLB PITCHERS

Kevin Lynch

Larry Nguyen

Charlotte Suarez

Alejandro Zamudio

Team 7

TODAY'S LINEUP

PRESENTATION OVERVIEW

01 Introduction

02 The Data

03 Problem Setup

04 Models

05 Results

06 Conclusions



INTRODUCTION

01 Background

- In 2022 the average length of an MLB game was 3 hours and 4 minutes
- 11th consecutive year with an average over 3 hours
- Pitch clock was introduced at the start of the 2023 season
- In 2023 the average length of a game was 2 hours and 36 minutes, a record low

02 Motivation

Evaluate how the introduction of the pitch clock has influenced pitcher performance, using our data to identify if and in which aspects of the game this impact is most evident

03 Focus and Approach

- Analyze which performance metrics have improved or declined
- Identify specific positive and negative indicators of performance changes
- Use predictive insights to guide decisions about pitch clock policy and gameplay impact



THE DATA



01 Statcast

- Introduced in 2014 as a way of tracking in-game metrics
- A single database for all MLB game data
- Pitch-by-pitch tracking allows for in-game analysis of performance

02 Database

- Pitch-by-Pitch Event Tracking: Every pitch and its result of every game in 2022 and 2023
- Yearly Aggregated Statistics and more advanced statistical metrics used for baseline statistics of league averages and trends.

03 Limitations

- Does not report pitch clock violations
- The pitch clock had only been implemented for one complete season
- Circumstances, situations and momentum are important, unreported, factors in the game

PRE-PROCESSING



01 PyBaseball

- Scrape yearly and single-game data

02 Renaming

- Altered column names
- Renamed NA events as a pitch thrown
- Removed non-pitch related features

03 Filtering

- Removed 2022 post season
- Removed 2023 spring training
- Removed pitchers who threw less than 30 innings each season

04 Separation

- Pre-Pitch/Post-Pitch Clock
- 2022/23 Batters
- 2022/23 Pitchers
- 2022/23 Advanced Pitching Metrics

PROBLEM SETUP

Pitch Specific Data

- Features: Pitch Type, Pitch Velocity, Exit Velocity, ...
- To inform us how a pitcher is doing pitch-to-pitch and batter-to-batter, tries to quantify “momentum”

Single Game Data

- Features: Strikes, Walks, Hits Allowed, ...
- To inform us how a pitcher is doing game-to-game and in how many outings he struggled

Yearly Aggregated Data

- Features: Wins Above Replacement (WAR), Earned Run Average (ERA), ...
- To inform us how well a pitcher performed in a season



RANDOM FORREST CLASSIFIER

Trained On

- ERA
- Average launch angle of hits allowed
- Hard hit percentage of balls put into play

Assumptions

- All input features utilized are relevant predictors of performance
- The model can use it to capture underlying aspects
- Features are well distributed and independent

With the classifier focused on determining which features are key to classifying data, we determined it would be effective with use of our in-game metrics. The model aims to classify improvement or decline in performance based on the features provided. Using WAR to create the classification variable, we aim to allow the model to identify which features play higher roles in classifying performance while observing its classification of player performance of the 2023 season. Predictions of improvement or regression in performance, will highlight any large difference between seasons.

LOGISTIC REGRESSION MODEL

Trained On

- Strike percentage
- Walk percentage
- Other pitch-specific metrics from our data of players in 2023

Assumptions

- The input features chosen will be able to represent our performance classifier effectively and highlight how poor or better a pitcher performed against league averages
- No Multicollinearity
- Observations Remain Independent

A classifier was generated for each pitcher for how well or poorly they performed by using ERA-, a normalized statistic meant to rate players based on their ERA compared to the rest of the league. Using related pitch specific metrics, the regression can indicate to us how well these aspects correlate to good or poor performance.

It is important to note that *many of these features used have the potential to be dependent on others*. Principal Component Analysis has been used to ensure multicollinearity does not exist and affect the observations of our classifier.

K-MEANS CLUSTERING

Trained On

- Walks and Hits Per Inning (WHIP)
- ERA
- WAR

Assumptions

- Clustering will highlight differences in performance
- The model can capture underlying aspects
- No Multicollinearity
- Observations Remain Independent

K-Means Clustering was used to observe significant changes in overall quality of performances. With an emphasis on our yearly aggregated statistics for each player, the model generates clusters and observes similarities in performance. By splitting our data of pitchers into different clusters, we can observe the differing numbers in their performance statistics. With the three statistics serving as the baseline, we can utilize the placement of clusters to see how each pitcher performed in terms of yearly performance.

MODEL EVALUATION

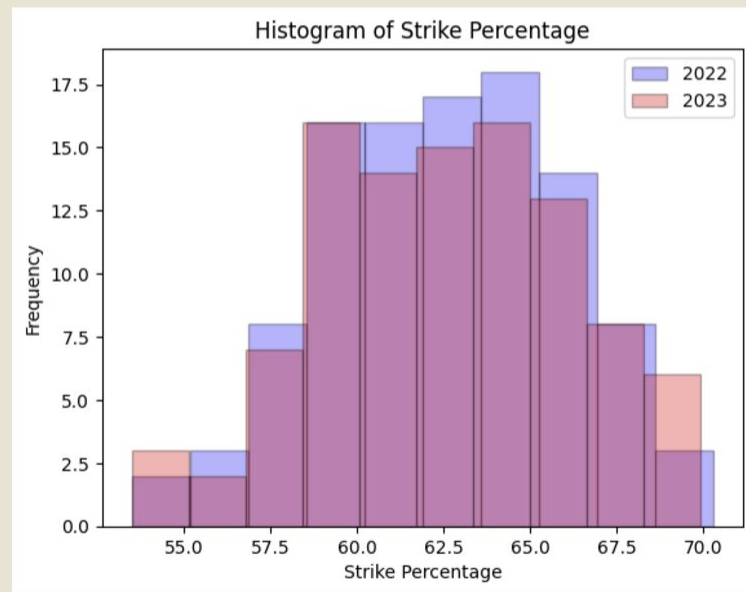
Our data presents various metrics, variables, and aspects to consider when evaluating performance. We split our data into multiple categories to focus on which metrics provide us with the most accurate lens into performance.

- All three models went through cross-validation to tune hyperparameters and ensure as much efficiency during performance.
- The datasets underwent a categorical encoding process in order to ensure our classification models could function correctly with these specific features.
- PCA was used to avoid multicollinearity.
- All data in was checked in order to ensure it was in correct formats and no missing values were present.

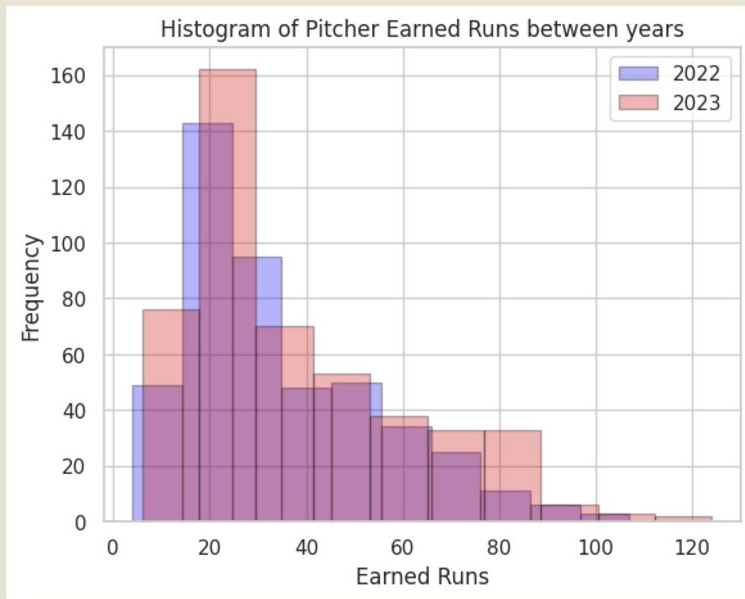


RESULTS

The histogram of strike percentage reveals that pitchers threw a higher percentage of strikes in 2022 compared to 2023. This shift suggests that pitchers may have faced challenges maintaining precision in individual games under the pitch clock in 2023. The decrease in strike percentage could reflect adjustments to the faster pace, potentially leading to less consistency in locating pitches within the strike zone. This finding underscores how the pitch clock may introduce variability in game-by-game performance, even if season-long statistics like ERA or pitch velocities remain relatively stable.



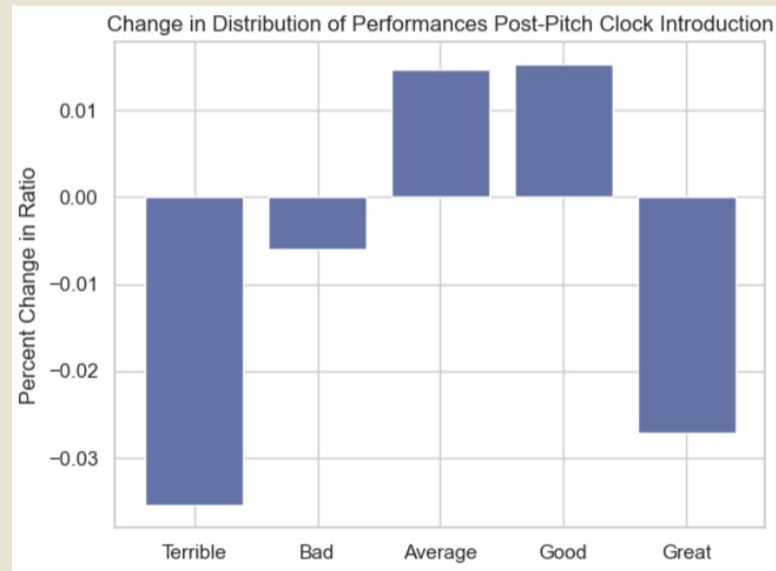
RESULTS



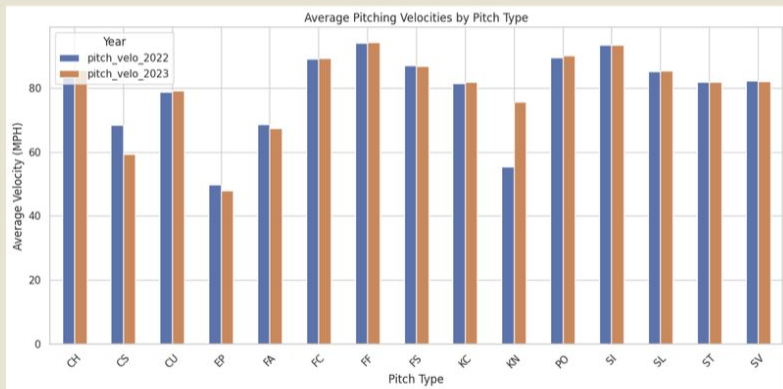
The histogram of ERA demonstrates a noticeable rightward shift and higher distribution in 2023 compared to 2022. The shift suggests that the faster pace of play may have had a compounding effect, making it harder for pitchers to recover during innings and leading to more runs scored overall. While ERA is a season-long statistic, this shift reflects the aggregate impact of subtle struggles observed in individual games, such as the decrease in strike percentage. These results highlight how systemic changes like the pitch clock can subtly elevate the baseline of performance metrics across an entire season.

RESULTS

As seen across the change in distributions with the introduction of the pitch clock, there was a sharp decrease in terrible and great pitchers. As a collective, the pitchers averaged out. This led to an increase in the distribution of average and good pitchers. The lack of time most likely makes it harder for the batters to adjust to the pitches, thus making it easier for terrible pitchers to perform average. Although the decrease in great pitchers can be attributed to the lack of time on the pitching side, not allowing for proper recovery for the pitchers or for them to think through more of their pitches.



RESULTS



For the most part, the average pitch velocity did not change substantially between seasons. The one pitch type it did, Knuckle Balls (KN), only jumped so high because there were very few between datasets. The steadiness of pitch velocity between seasons serves to indicate that even amid broader shifts in pitcher performance distribution, certain aspects, like the fundamental mechanics of pitch speed, remain consistent. It suggests that while the pitch clock can influence overall game dynamics and outcomes, it does not necessarily disrupt every technical facet of a pitcher's performance.

CONCLUSIONS



Pitchers

- Collectively, pitchers averaged out
- Those who performed very well and very poorly, lost their uniqueness
- 57 of 96 pitchers regressed in the random forest



Specific Metrics

- Launch angle increased
- Exit velocity increased
- Pitch velocity remained consistent
- Pitch type usage remained consistent
- Strike, walk, and in play ball rates remained consistent



Overall Game Change

- Pitchers experienced greater change game-to-game than season wide
- The pitchers did not adapt to the pitch clock at the rate the batters did, thus resulting in average yearly stats and higher launch angle and velocity



THANK YOU!

CATCH YOU NEXT TIME!
QUESTIONS?

Kevin Lynch	10
Larry Nguyen	10
Charlotte Suarez	10
Alejandro Zamudio	10